Problem Chosen	2026 HiMCM Summary Sheet	Team Control Number
A		18213

Choosing a City for High-School Life with Incomplete Evidence

Summary

We compare sixteen cities for high-school life across ten indicators. Under baseline weights, eight indicators at 0.1125 and two facility indicators at 0.05, the recommendation is Oslo, with score interval 56.43-63.90 and maximum regret 15.75, ahead of Zurich at 16.94 and Singapore at 16.95. Equal weights select Zurich. All sixteen cities remain possible champions, so this is a conservative starting point rather than a claim that one city dominates.

Official records supplement 25 originally missing city–indicator entries without replacing team values. The mean score-interval width falls from 18.81077 to 15.77678, a 16.13% reduction. The interval revision retains 48 uncertain entries and changes 14 fixed-weight ranks from midpoint completion; separate interior completions let each city lead.

Withholding Zurich's homicide point retains Oslo under baseline weights and selects Oslo under equal weights. Withdrawing its work-commute proxy selects Oslo under both; withdrawing Oslo's commute proxy alone selects Zurich under both. Withdrawing all twelve means selects Singapore and Zurich, respectively. Both preference profiles select Oslo. In 10,000 baseline weight perturbations at Dirichlet concentration 100, Zurich is recommended in 25.48% of draws, compared with 57.38% for Oslo and 17.10% for Singapore. These frequencies are conditional on the chosen weight distribution.

Under uniform ten-per-cent development changes, Oslo is selected in all three scenarios under baseline weights; equal weights select Oslo, Zurich and Oslo. City-specific shocks are illustrative, not forecasts. Country, district and age-group cases distinguish facility supply from student access and broad observed groups from unsupported age-specific preferences.

Keywords: high-school opportunities; TOPSIS; interval evidence; maximum regret; preference sensitivity.

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1 Introduction and decision scope

1.1 What makes a city suitable for high-school life?

A useful comparison connects a student's school day with the opportunities and constraints around it. Safety and air quality affect the daily setting; teaching resources and nearby higher education describe educational supply; housing, living costs, transport, travel, broadband and swimming facilities describe household pressures and everyday resources. We define "best" as one that minimizes its worst-case shortfall behind any rival within the retained ranges. That definition favors a city that is hard to beat over one that merely looks strong once missing values are filled in. It provides a starting recommendation for the compared cities, rather than a claim about every student's experience.

The comparison uses the frozen team matrix of sixteen cities, one per country, exceeding the ten-city minimum and spanning different resource and cost settings. This inherited set is not a representative global sample. Findings concern these candidates; broader claims require an explicit sampling design and additional comparable observations.

1.2 Criterion selection and contribution

The ten indicators preserve the agreed team framework and balance education with the conditions that make resources usable in daily life. Their directions and anchors appear in Table 1. Counts and city averages serve as scoped supply or setting measures. They do not directly measure friendships, student wellbeing, school quality, language fit, discrimination, cultural participation or individual access. Those outcomes are relevant to a family's choice; comparable student-level observations are needed to integrate them without inventing measurements or double-counting existing proxies.

Our central contribution is a city decision that retains incomplete evidence. We improve source precision, compare a point-completion benchmark with an interval revision, and examine preferences and future conditions without changing the primary indicator set. TOPSIS provides one main quality score; AHP structures stated priorities, Monte Carlo measures specified perturbations, and VIKOR supplies a focused method comparison. The extensions test the recommendation rather than vote among an expanding set of ranking models.

AI assistance supported research retrieval, calculations, code and prose preparation, including Exa source retrieval and extraction [19, 20]. The appended use report records the known assistance and the outstanding documentation required before submission.

2 Assumptions and their role

1. **Comparability follows the retained scope.** Original values and population denominators remain fixed. Each supplement keeps its provider, year, city boundary and educational population. Mixed-year densities are conditional comparisons; they do not create a same-year census.

2. **Utilities are monotone on fixed scales.** More resources and less cost or burden improve their respective utilities. Declared anchors set the scale; they are neither estimated preferences nor empirical optimum values.
3. **The score permits compensation.** Weighted distances combine criteria in a common utility space. The model therefore allows stronger results on some criteria to offset weaker results on others. A family requiring a hard safety or eligibility threshold should apply it separately before choosing.
4. **The evidence set is rectangular.** Uncertain utilities vary independently within retained intervals. This conservative closure omits unknown cross-criterion and cross-city restrictions. Its possible champions and regret bounds are conditional on this set; they are not population probabilities.
5. **Extensions hold the comparison basis fixed.** Scenario changes retain population denominators, anchors and the indicated weight rule. They are conditional experiments with no fitted future date or representative age-specific preference distribution.

3 Notation

Symbol	Meaning
M, N	Sixteen cities and ten indicators
w_j, a_j	Normalized criterion weight and fixed utility anchor
u_{ij}	Utility of indicator j for city i, on $[0, 1]$
ℓ_{ij}, h_{ij}	Retained lower and upper utility bounds
L_i, U_i	Lower and upper TOPSIS score, on $[0, 100]$
$R_i, \mathcal{C}$	Maximum regret and set of possible champions
κ	Concentration of a declared Dirichlet weight distribution
r	Raw-value multiplier in a development scenario

4 Data collection and comparability

4.1 A traceable city–indicator matrix

We compare $M = 16$ cities on $N = 10$ indicators covering safety, environment, household costs, education and everyday resources. The initial matrix is the team-supplied city table [1]. We retain its existing values, populations, reference years and source flags as the agreed input, then supplement only fields empty in that original table. Some supplied values lack original source URLs. The supplied Oslo price-to-income ratio of 9.8 equals the value in an archived Numbeo Current table; its annual 2026 edition gives 11.6 [63]. Neither the original extraction nor municipal-boundary equivalence is established. The team table remains the provenance of these inputs; official supplements do not retrospectively validate them. Table 1 states what each variable measures.

The nominal reference year is 2025; supplements retain their actual years and supplied population denominators. Matching Eurostat records give Dublin's 28.8 minutes for 2011,

Table 1: Retained indicators, units and reference anchors. Supplied values are retained; supplements have explicit source scope.

Indicator	Measurement and interpretation	Anchor
Homicide	Deaths per 100,000; cost; supplied city proxy	2
$PM_{2.5}$	$\mu g/m^3$; cost; supplied concentration proxy	15
Living costs	Supplied living-cost index; cost	100
Housing	Supplied price-to-income ratio; cost	10
School resources	Pupils per teacher or explicitly scoped FTE; cost	15
Higher education	Institutions per 100,000; benefit; supply proxy	1
Transport stops	Unique stops per 100,000; benefit; partial inventories	50
Travel	Duration in minutes; $\exp(-t/30)$; commute or stage proxy	30 min
Broadband	Supplied Mbps; benefit; city/general-use proxy	100
Swimming pools	Venues per 100,000; benefit; supply proxy	1

Zurich's 39.9 for 2024, and Oslo's 31.0 for 2009 as journeys to work, currently labelled "greater city"; Zurich's value is flagged estimated [56]. These matches establish neither the team's source lineage nor the historical boundary. We retain the supplied means, years and flags as conditional general-commute proxies. They do not identify high-school journeys. Mixed-year densities and historical inventories remain conditional; withdrawal scenarios examine their role in the decision.

We use official statistical tables, operator inventories and municipal geodata to fill missing fields. For each admitted supplement, the archive records the provider, URL, year, geographic and population scope, units, extraction rule and file hash. Original PDFs and API responses are retained separately from derived tables. Search summaries locate evidence; numerical admission requires the underlying record or an explicitly identified provider capture. Values are converted to compatible units before utility calculation: for example, the supplied higher-education density per million is divided by ten to obtain the per-100,000 scale. Mode share is not converted into physical stop density.

4.2 Collection problems and their resolution

Count city facilities, not platforms or entire networks. Entur's 9 November 2026 registry export supplies 626 Oslo bus stop groups under the source-defined pure on-street bus subset and native parent grouping [6]. The questioned Tveten bru entry is excluded as a precaution, without a confirmed closure. The 2026 inventory uses the supplied 2025 population of 727,797; it is not a 2025 count or a count of all currently served stops. The alternative 829 mode-bearing registry groups use a broader counting definition; they are not added to the 626, which replaces the earlier 89-station metro subset. Zurich's 2025 VBZ report distinguishes 446 city stops from 235 outside the city [5, 7]. Helsinki's November 2026 operator snapshot separates 19 Helsinki stations from 11 in Espoo [8].

For a verified subset count c, fixed population P and anchor a, we use $d = 10^5 c/P$ and utility $[d/(a+d), 1]$, leaving uncounted providers unknown. Madrid's May 2026 register has 293 records, of which 242 carry city code 28079. Omitting two unnamed records without lines and consolidating native station identifiers gives 191 complexes [24, 25]. Vienna's 10 November 2026 GTFS subset replaces 99 metro areas with 378 DIVA areas, qualified by public A-route regular calls, exact name/source-coordinate matches and all-position city tests [29, 30].

Native identifiers, verified aliases and operator interchange maps ensure that a transfer station is counted once rather than once per line. Prague retains 57 current metro complexes; Flora's closure explains its omission from the 58 distinct complexes in the 2025 report, whereas Florenc's partial accessibility restriction does not close the whole complex [31, 32, 33]. Berlin's nine BVG U-Bahn profiles consolidate 131 zone-A appearances into 100 complexes. Zone A extends through the S-Bahn ring; the retained U-Bahn subset leaves outer stations and other modes uncounted [34]. Milan's municipal layer, distinguished from its entrance layer, selects records with both the Milano name and code 015146, yielding 104 stations/hubs after five map-supported interchange consolidations; proximity alone does not merge stations [50]. Tokyo's Oedo ticket-service directory lists 38 distinct stations in the 23 wards; machine flags are not added. September 2026 updates describe the catalogue; its inventory date is unstated [57]. Warsaw's 21 M1 and 18 M2 entries yield 38 complexes after merging Świętokrzyska. Its 2024 map distinguishes planned stations [58]; a 2025 fire report locates the lines within Warsaw [59]. Counts use Tokyo's 2026 and Warsaw's 2024 populations.

Amsterdam's 39 provider-grouped Metro locations give 35 inside both municipal 2026 and CBS 2025 boundaries; a declared 100-metre nominal boundary screen retains 32 [54, 55]. The screen defines a subset, not an accuracy guarantee. The 2026 metadata leave its inventory date unspecified. The retained 2025 population of 941,927 gives a conditional lower bound, leaving other transport supply unknown.

London's inventory mixes platforms and entrances with stations. From 272 station-level records, 62 have exact fare-zone value 1; it retains 59 stations/hubs after two conservative omissions and Paddington hub consolidation; representative coordinates lie within the ONS December 2025 London-region boundary [37, 38]. Singapore's LTA GeoJSON contributes 5,205 distinct nonempty bus-stop codes, without distance clustering [42]; Stockholm's 2025 city–police report identifies thirteen metro stations in two districts [49].

Keep inventory dates, status and venue units explicit. Milan's layer is dated 16 October 2024; Paris's 244 intra-muros metro stations come from a March 2022 atlas using RATP 2021 and Apur 2022 sources [50, 10]. These historical lower-bound proxies are not relabelled as 2025 counts; a separate withdrawal restores the Paris supplement to unknown. Vienna, Berlin and London use November 2026 inventories. All retain the supplied population denominators, including Paris's 2023, Dublin's 2022 and Singapore's 2024 values. Inventory membership does not establish continuous operation or adolescent access. Stockholm's November 2026 directory retains thirteen municipal swimming venues after excluding a renovation closure and counting its replacement once [9]. Tokyo retains five Nerima indoor venues at distinct addresses: Heiwadai's current closure excludes

it, while Miharadai's announced February–May 2027 closure is not a current closure [26, 27]. Named subsets include 25 Singapore venues in the ActiveSG directory and one OCBC Aquatic Centre at 7 Stadium Drive, Singapore 397632, counted once despite its multiple basins; operator pages were captured 10 November 2026 and their inventory date is unstated. The denominator is the 2024 population of 6,036,900. Twelve Zurich venues and seventeen Milan indoor venues under the June 2025 tariff are also retained [43, 45, 44, 47, 48]. The Singapore count does not establish opening or teen access. Four earlier pool-specific notices affect three ActiveSG venues: withholding them leaves 23 venues; also withholding OCBC leaves 22. At OCBC, competition-pool booking is for ages 16 and above, training-pool booking is for all ages, and supervisory rules apply for ages 15 and below. Individual per-entry prices are offered; normal hours are 07:00–21:30 Monday and Wednesday–Sunday. Tuesday maintenance, the 30 October–1 November 2026 event and 5–28 December competition-pool closure with limited training availability constrain access [44]. Zurich includes five distinct school pools with limited public sessions, some women/girls-only, closed during school holidays and non-school days; adult-only Kügeliloo is excluded. Venue supply does not establish adolescent access; other providers remain unknown.

Match school stages and teaching units before forming ratios. Prague's 2024/25 totals are 80,073 secondary-school pupils and 6,877.1 teacher full-time equivalents (FTE), including lower-gymnasium years. Subtracting the matching 8,748 pupils and 737.8 FTE gives 71,325 upper-stage pupils and a rounding envelope of $[6139.2, 6139.4]$ teacher FTE, including adult and non-daytime study modes [11]. Stockholm's city-school ratio covers upper-secondary years 1–3, all ownership types and the 15 October reference date. Its API value 16.42091405 is retained as $[16.35, 16.45]$: a declared precision convention, not a measurement-error confidence interval [12]. Warsaw's 2024/25 yearbook gives 119,740 pupils across six youth school routes by school location and all ownership types. Unknown youth allocation within adult-inclusive general-school FTE gives 2,490–8,144 FTE; outward one-FTE guards on four published cells widen this to 2,487–8,148 [46]. The guards encode a unit-rounding assumption. These allocation and rounding bounds describe school resources, not a resident age-14–18 population or a statistical confidence interval.

Some published denominators fail these checks. Henri-IV in Paris reports 794 lycée pupils and 153.65 institution FTE for 2025, without upper-stage allocation or complete city coverage; a 2026 post-baccalaureate directory flag cannot recover 2025 workloads [35, 36]. Dublin's school-roll payroll FTE are unpublished [28]; its November 2025 City Of Dublin ETB post-primary vocational subgroup has 411.8 payroll FTE against 376.1 allocated posts, so allocations are not a payroll ceiling [51]. Finland's national FTE combines register, salary and private-provider estimation; without a verified one-sided error envelope it is not a guaranteed Helsinki teacher upper bound [52]. All three city PTRs remain unknown.

Preserve incompatible rate units and supplied values. Amsterdam's CBS sources give ten provisional victims in 2022 and 68 across 2018–2022, neither matching the supplied numerator 340 [40]. Zurich's PKS lists one completed offence in 2024 and two in 2025;

offences count criminal acts independently of victims, and the submitted rate 0.14 still lacks a verified denominator [41]. Both original entries remain, with their contribution tested in Section 5.5; we do not manufacture replacement rates from incompatible units.

Retain unobserved travellers and failed retrievals as uncertainty. Singapore's initial census-PDF endpoint returned 404; the official Parliament mirror and API supplied the matching report and table [14]. Duration-bin utility uses endpoints, no midpoints or artificial open-tail cap. Weighted enrolment estimates exceed travel coverage by 1,811 secondary, 989 post-secondary non-tertiary and 1,635 polytechnic students. These are weighted survey estimates, not complete headcounts. Uncovered mass retains $[0, 1]$ utility, and unknown stage shares use the interval convex hull. The resulting $[0.21048, 0.61884]$ is a 2020 school-stage proxy, not an identified 2025 interval for ages 14–18. Dublin's stop-download 403 yielded no catalogue-inferred count; a later official map from the 2023 plan, issued in 2024, supports nineteen existing city-centre tram stops [15]. Warsaw's recovered 2022 survey reports a rounded 27-minute school trip for a city subgroup with sample base 22, but covers residents aged 15 and above without an upper-secondary or matched 14–18 denominator [39]. Its travel utility remains unknown.

Figure 1 locates the remaining uncertainty. The 160 cells comprise 112 points, 37 partial intervals and 11 fully unknown entries: eight teacher-ratio and three travel entries. The original table had 112 points, 12 partial intervals and 36 unknowns. Evidence for twenty-five distinct city–indicator cells fills original gaps while preserving supplied measurements. Facility supply, commute proxies and school resources remain distinct from adolescent access, school travel and learning quality.

5 Establishing the baseline evaluation model

5.1 Determining indicator weights

We retain eight main indicators at weight 0.1125 each and two auxiliary facility indicators, transport stops and pools, at 0.05 each, so $\sum_j w_j = 1$. These weights express a transparent modelling choice. They are not estimated from a student survey. Equal weights provide a separate reference. TOPSIS is the primary score; VIKOR provides a focused method comparison. AHP represents explicit preference scenarios [2], and Monte Carlo measures sensitivity to declared distributions. These extensions retain the same criteria rather than introduce unrelated ranking models.

For preference multipliers $m_j > 0$, we use

$$w'_j = \frac{m_j w_j}{\sum_k w_k m_k}, \qquad A_{jk} = \frac{w'_j}{w'_k}. \tag{1}$$

The principal eigenvector of A recovers w'. With $\lambda_{\max}$ its largest eigenvalue, $CI = (\lambda_{\max} - N)/(N-1)$ and $CR = CI/RI_N$. Because this matrix is built from exact ratios, $\lambda_{\max} = N$ and $CR = 0$ up to numerical precision. This verifies internal consistency; it does not establish that these are students' actual preferences.

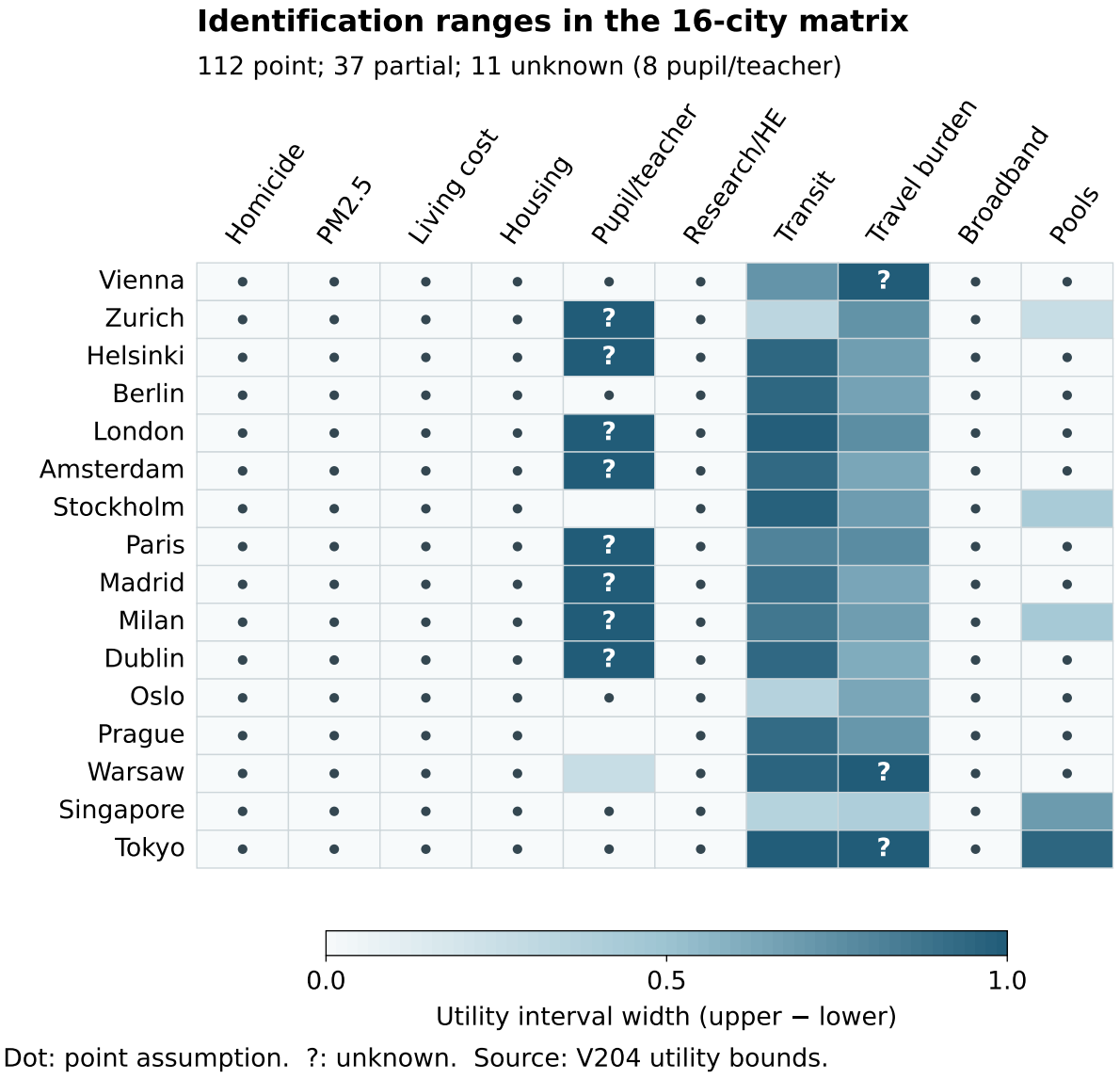


Figure 1: Identification ranges in the retained sixteen-city, ten-indicator matrix. Eight of the eleven fully unknown entries concern upper-secondary pupil–teacher ratios. Shading shows utility upper minus lower bounds; dots mark retained point assumptions and question marks identify $[0, 1]$ entries. Point assumptions include submitted values and do not certify independent validation or zero measurement error. These are identification bounds under the retained source rules, not confidence intervals. Travel entries retain school-travel and general-commute proxy scopes. Cities follow the input order. Source: V204 utility bounds.

5.2 TOPSIS scoring calculation

Fixed-anchor TOPSIS [3] gives benefit and cost utilities, respectively:

$$u^+(x) = \frac{x}{a+x}, \qquad u^-(x) = \frac{a}{a+x}. \tag{2}$$

Each anchor in Table 1 is a declared reference scale, not an estimated optimum. Travel uses $q(t) = e^{-t/30}$ for duration t in minutes. When only mean duration $\bar{t}$ is known, Jensen's inequality gives $\mathbb{E}[q(t)] \in [e^{-\bar{t}/30}, 1]$. Fully missing utilities remain $[0, 1]$.

For utility vector u_i, weighted distances from fixed worst and best points are

$$D_i^- = \sqrt{\sum_j w_j u_{ij}^2}, \qquad D_i^+ = \sqrt{\sum_j w_j (1-u_{ij})^2}, \qquad S_i = 100\frac{D_i^-}{D_i^- + D_i^+}. \tag{3}$$

Weights enter the squared distances once. Since S_i increases with every utility, endpoints ℓ_i, h_i give $L_i = S(\ell_i)$ and $U_i = S(h_i)$. Under the retained rectangular uncertainty set, define

$$R_i = \max\{0, \max_{k \neq i} U_k - L_i\}, \qquad i^* \in \arg\min_i R_i, \tag{4}$$

$$\mathcal{C} = \{i : U_i \geq \max_{k \neq i} L_k\}. \tag{5}$$

Maximum regret R_i measures how far the selected city could trail another city. Set $\mathcal{C}$ retains every city that could still be first. The maximum excludes the selected city itself. These conditional data bounds are not sampling confidence intervals, and a conservative recommendation does not identify a uniquely best city.

5.3 Scoring results and interpretation

Oslo minimizes maximum regret, followed by Zurich and Singapore (Table 2). Oslo's lower score bound is 56.43 and its maximum regret is 15.75; Zurich's corresponding values are 53.97 and 16.94. All sixteen cities remain possible champions, and none is necessary. These rankings express a conservative recommendation under the stated criteria and intervals.

Holding weights and formulas fixed, evidence for twenty-five distinct city–indicator cells narrows the average fixed-weight score interval from 18.81077 to 15.77678, a 16.13% reduction. Vienna's 2026 scheduled-stop subset replaces 99 metro areas with 378 DIVA areas, reducing fixed-weight mean width from 15.82233 to 15.77678; population remains 2025. The revised Oslo transport subset changes the baseline conservative recommendation. Withdrawing Singapore's historical school-stage travel proxy retains Oslo under baseline weights.

At the same illustrative midpoint utilities and weights, VIKOR equally balances total and largest shortfalls using sample best/worst values [4]. At midpoint utilities, TOPSIS selects Zurich. VIKOR's lowest Q is Dublin with fixed weights ($\tau_b = 0.483$) and Oslo with equal weights ($\tau_b = 0.650$). Dublin meets both acceptance conditions; Oslo fails stability and is not an accepted single compromise. This diagnostic reveals method dependence; Equation 4 retains TOPSIS and the evidence intervals.

Table 2: Conservative baseline decision ranks with unchanged weights.

Rank	City	Score interval	Regret	Rank	City	Score interval	Regret
1	Oslo	56.43–63.90	15.75	9	Paris	49.11–69.04	23.08
2	Zurich	53.97–72.18	16.94	10	Madrid	49.00–68.22	23.18
3	Singapore	55.23–63.53	16.95	11	Vienna	48.25–61.03	23.93
4	Stockholm	52.12–64.54	20.06	12	Warsaw	47.64–63.99	24.54
5	Prague	51.97–62.78	20.21	13	Milan	46.07–65.99	26.11
6	Dublin	51.13–70.90	21.05	14	London	45.22–66.12	26.97
7	Berlin	50.73–61.54	21.45	15	Tokyo	43.98–61.12	28.20
8	Helsinki	49.41–69.84	22.78	16	Amsterdam	42.48–60.43	29.70

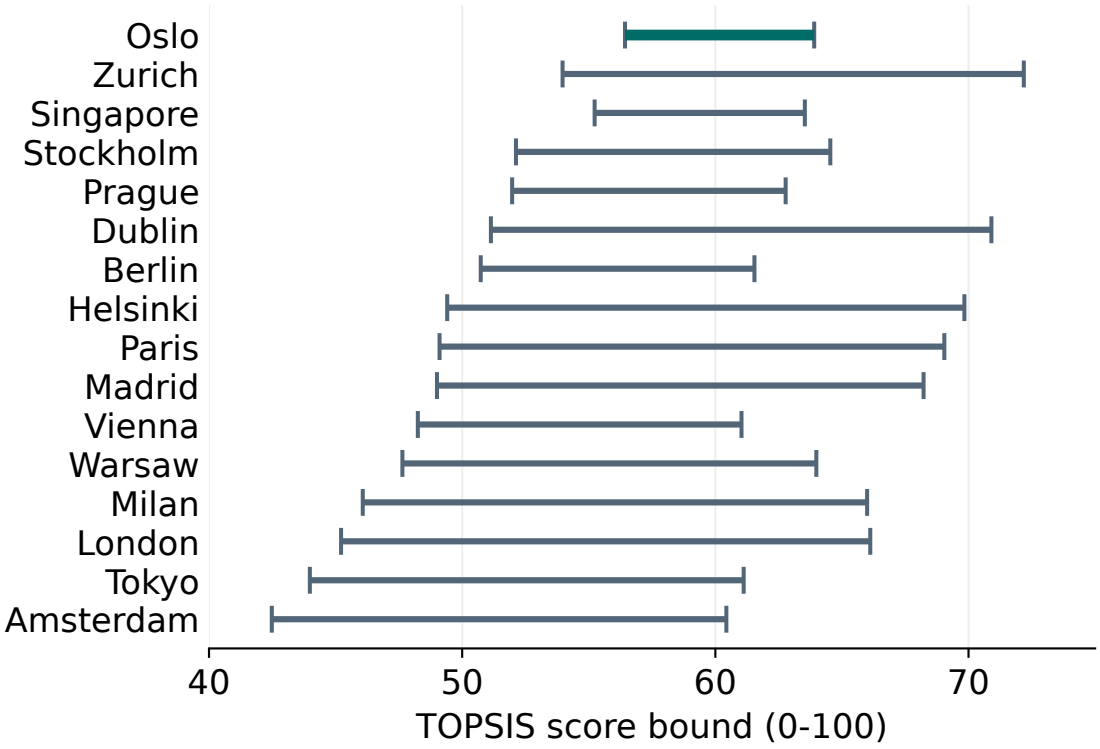


Figure 2: Current score bounds and conservative decision order for the sixteen cities. Intervals retain missing information; they are not confidence intervals. Source: frozen city matrix and admitted supplements.

5.4 A significant revision: retaining uncertain evidence

A controlled comparison shows why evidence bounds belong in the decision. On the same 160 utility entries, compare the midpoint completion $z_{ij} = (\ell_{ij} + h_{ij})/2$ with the retained-interval formulation in Equation 4. Both use identical weights, anchors and TOPSIS scores. The midpoint is a declared diagnostic scenario, rather than an estimate of missing observations or a reconstruction of the team's historical first model.

The revision retains 48 non-point entries and selects the city with the smallest worst-case shortfall. Oslo is the fixed-weight recommendation, and 14 cities change decision rank under fixed weights; Kendall's $\tau_b = 0.600$. With equal weights, 14 ranks change and $\tau_b = 0.683$.

For each target city c, construct a separate interior completion: set every uncertain utility to $\ell_{ij} + \alpha_i(h_{ij} - \ell_{ij})$, with $\alpha_c = 0.95$ and $\alpha_i = 0.05$ for $i \neq c$. All 112 point values remain fixed and all 48 uncertain values lie strictly inside their intervals. Each target ranks first under both baseline and equal weights. Across the sixteen targets, the smallest lead is 2.941 score points under baseline weights and 2.222 under equal weights. These separate completions establish feasibility in the declared utility rectangle; they carry no estimated probabilities and do not establish joint physical attainability.

Retaining intervals changes the decision order while preserving sixteen possible champions. The 16.13% reduction in interval width comes from admitted data, separately from this formulation comparison.

5.5 Sensitivity to the accepted input information

Point withdrawals replace selected point utilities by $[0, 1]$, keeping weights, supplements and the 48 non-point bounds fixed. Test pre-2020 dates, post-2025 dates, non-`ok` flags and their union, counting overlaps once.

Table 3: Recommendation after withdrawing selected submitted point information. Width is the mean baseline-weight score range on the 0–100 scale.

Withdrawn group	Cells	Baseline	Equal	Width
Retained input	0	Oslo	Zurich	15.78
Before 2020	13	Singapore	Zurich	20.88
After 2025	64	Zurich	Zurich	48.20
Non-ok flags	28	Oslo	Oslo	28.89
Union	95	Singapore	Singapore	68.48

Withdrawing 28 flagged points selects Oslo under both weights; removing 13 pre-2020 points selects Singapore or Zurich, respectively. Removing 64 post-2025 points selects Zurich under both and widens the baseline mean range from 15.78 to 48.20. The 95-point union selects Singapore under both. Dates and flags are screening criteria, not error probabilities.

Withdrawing the 626 bus subset and restoring the earlier 89 metro stations selects Zurich under both weight sets. Treating 829 registry groups as comparable units instead of the 626 bus subset selects Oslo under both weight sets. The broader definition does not establish separate physical facilities or current service. Withholding the two replacement bus groups, the Midtstuen bus group, or both retains Oslo under baseline and Zurich under equal weights.

Restoring Vienna's 99 metro areas preserves both choices. Seasonal, detour and callbus conditions remain unverified for individual trips. Withholding 36 selected witnesses on nine condition-bearing routes leaves 342 areas; Oslo and Zurich remain selected, without rematching.

Withholding Zurich's homicide point retains Oslo under baseline weights and selects Oslo under equal weights; withholding Amsterdam's retains Oslo and Zurich, respectively. Varying Zurich's rate between zero and five per 100,000 does not change the fixed-weight selection; under equal weights the tie occurs near 0.14981. These are hypothetical rates with other inputs fixed.

Withdrawing Oslo's commute proxy alone selects Zurich under both weights; all sixteen remain possible champions and none is necessary. Withdrawing Zurich's commute proxy, or both Zurich's and Dublin's proxies, selects Oslo under both weights; withdrawing all twelve means selects Singapore under baseline and Zurich under equal weights. Dublin alone leaves the baseline choices unchanged. Varying only Zurich's mean gives ties at 28.10 minutes under baseline weights and 40.50 under equal weights: Zurich is selected below each tie and excluded above. At the retained 39.9 minutes, the baseline selection is Oslo.

Jointly vary each of the twelve means t_i independently in $[(1-\delta)t_i, (1+\delta)t_i]$, $0 \le \delta \le 1$, retaining all other entries. Fixed upper scores make each regret increase with its own mean. The largest target mean and the other eleven smallest maximize the target's regret relative to the best rival. The critical radii are 15.6063% for baseline winner Oslo and 0.6483% for equal-weight winner Zurich. Below each boundary the respective winner uniquely minimizes regret throughout the box; at it the other can tie, and above it a counterexample exists. These conditional ranges are not measured errors or future probabilities.

6 Model revision and extension

6.1 Considering future changes

We examine three conditional changes while holding anchors and population denominators fixed. Resource expansion raises higher-education, stop and pool densities by 10%. Living pressure raises living costs, housing cost ratios and travel durations by 10%. Balanced improvement reduces pupil–teacher ratios, $PM_{2.5}$ and travel durations by 10%. A comparable multi-year panel is not available, so these changes have no estimated annual rate or forecast date.

For raw multiplier $r > 0$, the utility transformations are

$$T_r^+(u) = \frac{ru}{1-u+ru}, \qquad T_r^-(u) = \frac{u}{u+r(1-u)}. \tag{6}$$

These increasing maps propagate raw-scale changes through benefit and cost intervals. For known mean travel duration, the new Jensen interval is $[e^{-r\bar{t}/30}, 1]$. Singapore instead has bounds $[L_q, U_q]$ on expected utility. With $q = e^{-t/30}$ and $t' = rt$, Jensen and $0 \le q \le 1$

give

$$\mathbb{E}[q^r] \in \begin{cases} [L_q, U_q^r], & 0 < r < 1, \\ [L_q, U_q], & r = 1, \\ [L_q^r, U_q], & r > 1. \end{cases} \tag{7}$$

Raising both endpoints to r would generally misrepresent an expected utility. Equation 7 is an outer moment bound and need not be reversible. Fully missing travel remains unknown. Thus the extension accommodates the different information supplied by means and by travel distributions.

Fixed weights select Oslo in all three scenarios; equal weights select Oslo, Zurich and Oslo, respectively. With Singapore historical stage-travel withdrawn, these choices remain the same. All sixteen cities remain possible champions. We also solve a decision boundary: holding other cities fixed, proportionally reduce one city's pupil–teacher ratio, $PM_{2.5}$ and travel durations until its maximum regret ties that of the best other city. Oslo already minimizes regret at baseline. Zurich, Singapore and Stockholm reach a tie after hypothetical joint reductions in pupil–teacher ratio, $PM_{2.5}$ and travel duration of 14.149%, 23.331%, 43.003%, respectively. These are hypothetical thresholds. These thresholds describe the model response to a specified joint change; they do not establish policy feasibility or a uniquely identified champion. Berlin does not reach a tie within the tested 50% joint reduction.

For a separate future-shock simulation, each draw combines a common shock and city-specific shocks under a declared bounded distribution, with the criterion weights held fixed. Original data intervals are not randomly completed. Specifically, $C \sim U[0,1]$ is shared across cities and $I_i \sim U[0,1]$ varies by city, independently; $\delta_i = 0.05C + 0.05I_i$. The relevant raw multipliers are $1 + \delta_i$ or $1 - \delta_i$, and affected criteria within a city share that multiplier. With 10,000 draws per scenario, Oslo's fixed-weight recommendation frequencies are 100.00%, 100.00%, 100.00% for resource expansion, living pressure and balanced improvement respectively. Repeating the same shocks after withdrawing Singapore's historical stage-travel proxy gives 100.00%, 100.00%, 100.00% respectively. These paired calculations isolate dependence on that proxy while retaining the chosen shock distribution. These are conditional stress frequencies under an uncalibrated distribution. Weight perturbations below answer a separate question and are not blended silently into the future-shock frequencies.

6.2 Considering different preferences and development situations

Education priority doubles the weights of the two education indicators. The independent living scenario doubles travel and transport-stop weights. Equation 1 then normalizes all weights. Both education and independent-living priorities retain Oslo; their decision-rank τ_b values relative to baseline are 0.767 and 0.900. All sixteen cities remain possible champions. The 2026 Vienna scheduled-stop inventory changes the baseline evidence bounds. The resulting differences show how the definition of a desirable city affects the decision. They do not establish measured differences between fourteen- and eighteen-year-old students.

6.3 Perturbing weights and point assumptions

At each preference centre w', draw 10,000 weights from $w^{(b)} \sim \text{Dirichlet}(\kappa w')$ for each $\kappa \in \{20, 100, 500\}$. Recompute interval TOPSIS and maximum regret without filling missing values; ties share first- and top-three membership.

Table 4: Conditional recommendation frequencies (%) at concentration 100.

Preference centre	Zurich	Oslo	Singapore
Baseline	25.48	57.38	17.10
Education priority	2.65	71.59	24.78
Independent living	23.23	70.28	6.46

At concentration 100, mean decision ranks are Oslo 1.5680, Zurich 2.5167 and Singapore 2.4634. AHP consistency checks calculation, not representative student opinion.

For $n = 10{,}000$ independent, identically distributed draws per setting, Hoeffding's inequality and a union bound over $m = 2064$ exported means in 37 settings give 95% family-wise coverage with error at most $R\sqrt{\log(2m/0.05)/(2n)}$: 2.38 percentage points for frequencies ($R = 1$) and 0.357 ranks ($R = 15$) [53]. Difference intervals resolve 4,274 of 4,440 pairwise mean-rank comparisons; 166 remain unresolved. Under the declared simulation laws, 23 of 37 settings certify a unique smallest expected rank.

Separately, widen each of the 112 point utilities x to $[\max(0, x - \epsilon), \min(1, x + \epsilon)]$, retaining the other 48 intervals. At $\epsilon = 0.05$, minimax selects Oslo with fixed weights and Zurich with equal weights. Over $\epsilon \in [0, 0.2]$, fixed weights certify Oslo through 0.08135 and Zurich from 0.08136, with unresolved radii between; equal weights select Zurich throughout. This utility radius is not a raw-value percentage, calibrated error or confidence interval. Each choice applies to an expanded uncertainty box, not every exact dataset inside it; primary scores are unchanged.

For source-review priorities, withdraw each point separately to $[0, 1]$, holding the other 111 points and 48 existing intervals fixed. Recommendations switch in 10 of 112 fixed-weight cases (Oslo to Zurich) and 35 of 112 equal-weight cases (Zurich to Oslo); all cases retain 16 possible champions and no necessary champion. These ignorance scenarios establish neither measurement errors nor error probabilities.

6.4 Within-city fairness and growth stages

Citywide facilities per resident measure supply. Student access additionally depends on fees, opening hours, transport, disability access and location; identifying its distribution requires matched students and qualifying facilities.

An auxiliary spatial view retains residents and generic Learning POIs in Bruno et al.'s published grids [60]. In Oslo, 99.18% of retained residents live in cells with modelled mean walking time at most 30 minutes to 20 Learning POIs; Zurich's share is 99.97% (Figure 3). These grid means describe neither student trips nor access to the counted pools; they remain separate from the primary scores.

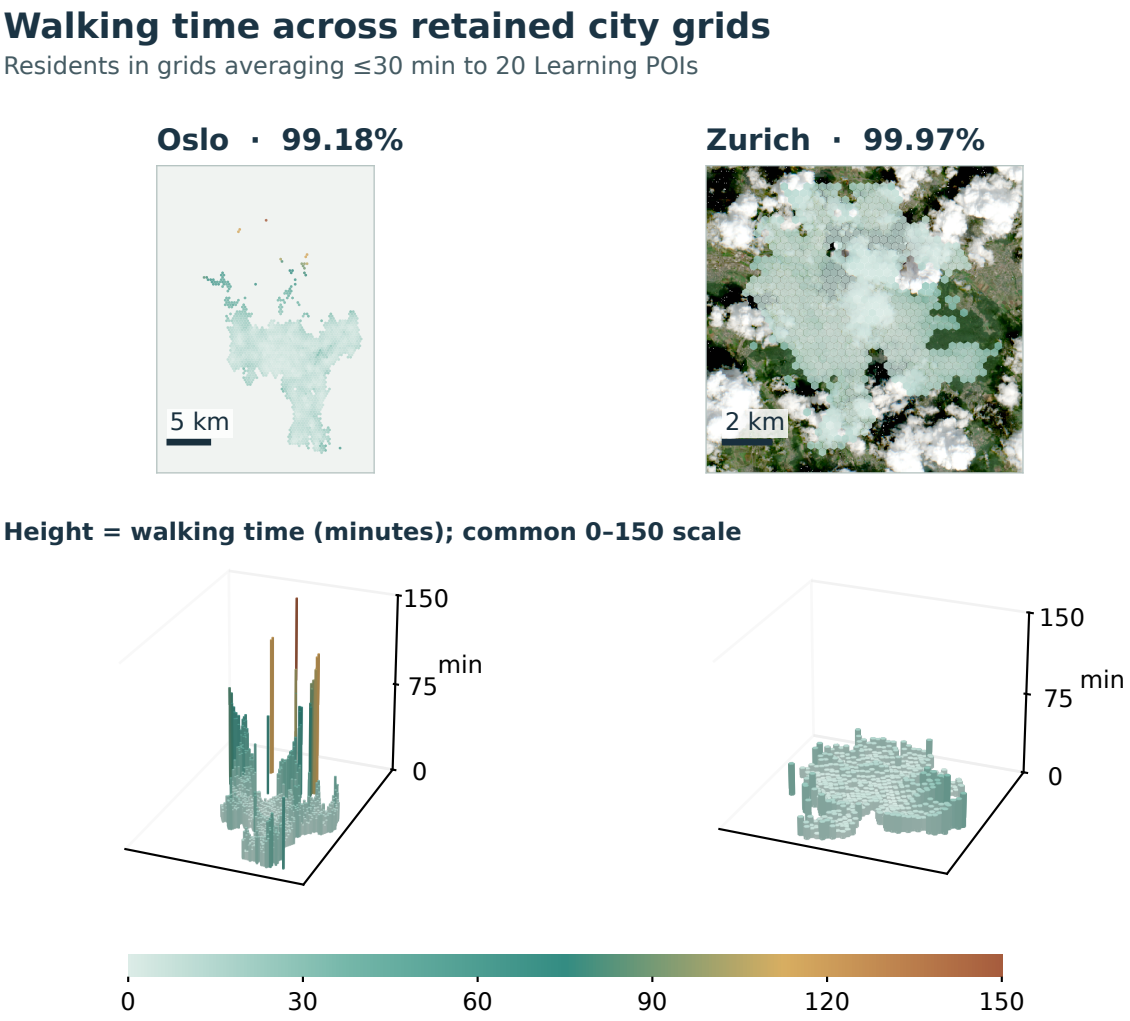


Figure 3: Modelled walking time to 20 generic Learning POIs in retained resident cells. Colours and heights use a common 0–150 minute scale; height is not terrain. Percentages weight cells with mean time at most 30 minutes by retained residents. Grids were captured on 10 November (Oslo) and 7 November 2026 (Zurich); input years are unstated and boundaries differ [60]. Oslo's retained 686,862 residents differ from the primary population of 727,797. Three Oslo NASA HLS requests were transparent, so its backdrop is geometry only. Zurich's requested 27 June 2024 background is cloudy; the granule date is unverified [61]. Neither panel measures student, high-school or pool access.

A separate 2025 Shanghai school-location case covers 16 districts, 235,542 regular senior-secondary pupils and 23,999 full-time teachers. Pooled resource utility is 0.604481 and the pupil-weighted district mean is 0.603267. The bottom 20% of pupil mass, with fractional boundary inclusion, has mean 0.580954, a gap of 0.022313. These district-homogeneous resource utilities measure neither individual teacher exposure nor resident access [16].

For districts d with pupil counts n_d and pupil–teacher ratios r_d, the pupil-weighted utility is $\bar{u} = \sum_d n_d \, 15/(15 + r_d)/\sum_d n_d$. Its difference from pooled utility reflects aggregation; school-location data establish neither resident travel access nor policy effects.

For Istanbul's 2012 high-school cohort aged 14–18, $L_{0.2}$ is the lowest-20% weighted mean travel utility, with fractional boundary inclusion, and μ the overall mean. Unknown utilities lie in $[0, 1]$; some membership is uncertain. Observed weight $W = 499950$ and all optional members give $N_{\max} = 681503$. The sharp joint bound $\min(\mu - L_{0.2}) = (W/N_{\max})(\mu_{\text{obs}} - L_{0.2,\text{obs}}) = 0.226686$ ("obs" denotes observed trips) is attained by including all optional members and setting unknown trips to 45 minutes. With fixed cohort, weights and missing-data rules, arbitrary utility changes to at most 5% or 20% of observed weighted mass give sharp minimum gaps 0.200134 or 0.129091, respectively. These declared fractional-mass budgets are not estimated error rates or fractions of students. The official-portal aggregate projection, rather than original CSV bytes, supports these historical feasibility results, not current facility use [17].

Istanbul's 2012 household survey provides current-high-school age groups 14–15 and 16–18. Their travel-utility marginal bounds overlap: $[0.35961, 0.55711]$ and $[0.29347, 0.59346]$. Unknown durations, optional members and shared unknown-age mass mean these extrema need not be jointly attainable. They do not identify today's exact ages fourteen and eighteen. Singapore's enrolment counts cannot be joined to resident travel cohorts without matched populations and reference days [17, 18].

Assigning shared unknown-age mass once gives the younger-minus-older mean travel utility joint closure $[-0.23330, 0.26325]$. Two finite-duration completions allocate shared mass to the younger group and use unknown utilities 0.05 and 0.95, yielding contrasts $+0.23842$ and -0.20848. Zero utility is a closure limit; the interior reversals are feasible constructions for the 2012 aggregates, not observed age effects.

In Oslo's January–March 2023 school survey, 66% of grade 8 and 41% of Vg3 respondents rated local provision of youth meeting places very or fairly good; 70% and 32%, respectively, reported sports-club activity at least once in the previous month [62]. Item/grade sample sizes are unreported; these cross-sectional comparisons establish neither exact ages, preferences, causes nor access and do not enter the primary scores.

Exact-age comparisons require matched resident utilities $u_{ij}^{(14)}, u_{ij}^{(18)}$ and separately elicited weights $w_j^{(14)}, w_j^{(18)}$; our analyst scenarios carry no age labels.

7 Comparing cities within the same country

The primary sample has one city per country. Frozen US and India cases use the same ten indicators and scoring formula, outside the sixteen-city ranking [21].

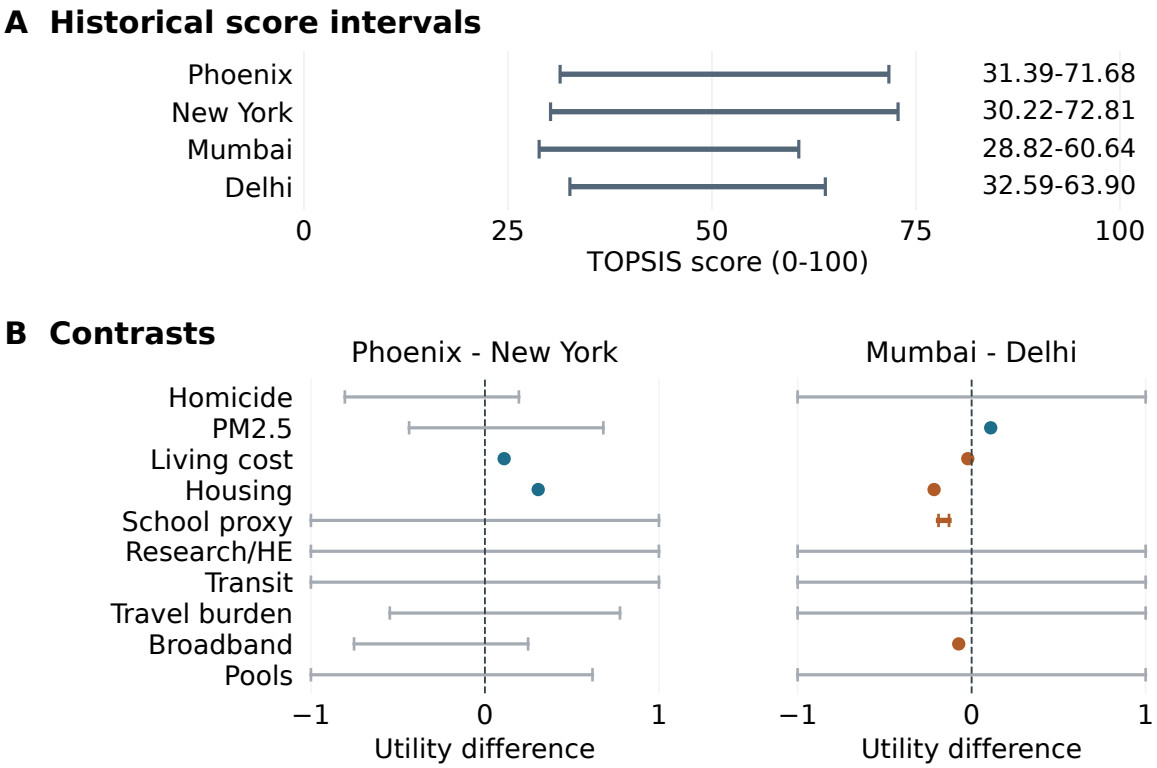


Figure 4: Frozen auxiliary V117 country cases, separate from the primary sixteen cities. Top: retained score intervals. Bottom: all twenty utility differences, first city minus second; seven exclude zero. Positive differences favour the first city; negative differences favour the second. Grey ranges leave direction unresolved. School resource retains 2024–25 school-location all-age XI–XII counts and nominal teacher-headcount allocation, not FTE. Living cost and housing use 2025 Numbeo aggregates; air uses 2021 WHO monitoring means; broadband uses the September 2026 index (July–September test window). Source years, boundaries and populations differ. Ranges are conditional identification bounds, not confidence intervals; neither pair identifies a necessary overall winner [21, 22, 23].

Phoenix has higher retained affordability utilities than New York. Mumbai has the higher $PM_{2.5}$ utility; Delhi leads in living-cost, housing, school-resource and broadband utilities. Both score pairs overlap. These historical city and education scopes identify neither a necessary winner nor a national causal effect.

8 Conclusion and decision use

Baseline weights recommend Oslo; equal weights recommend Zurich by minimax regret. All sixteen cities remain possible champions. Supplements narrow mean score-interval width by 16.13%.

A city choice built around a student's day

Our comparison of sixteen cities across ten aspects of high-school life gives families a starting point for weighing everyday trade-offs. Oslo is the cautious recommendation under baseline and education-priority weights; Zurich is selected when all ten measures receive equal weight. The priorities are analyst-designed, not drawn from a student survey. All sixteen cities remain possible leaders, so a family's choice can depend on the evidence and priorities it considers relevant.

How the comparison works

The measures cover safety, air quality, housing and living costs, teaching resources, higher education, transport stops, journey time, broadband and swimming facilities. Their units differ: a travel time and a pool count cannot be compared directly. The model puts each measure on a common reference scale, then combines them using the stated priorities. Missing or partial figures remain ranges instead of being filled with invented averages.

A single completed table could make one city look certain to win even when the missing information matters. We therefore compare each city's largest possible shortfall behind another city within the retained ranges. The cautious choice is the city with the smallest such shortfall. This gives families a way to discuss trade-offs while keeping uncertain information visible.

What new records change

Additional evidence narrows the average city-score range by about sixteen per cent while retaining the team's original values and scoring rules. For Oslo, a 2026 registry provides a source-defined subset of 626 on-street bus stop groups. It replaces the earlier subset of 89 metro stations, rather than being added to it; the comparison uses the supplied 2025 population. Restoring the earlier subset selects Zurich under both baseline and equal weights. This change shows how a source definition can affect a recommendation, and invites a closer look at what each city count includes.

The counts describe supply, not an individual student's access. A nearby stop may still require a suitable route, service time, fare and accessible connection. A pool count does not describe fees, opening hours or neighbourhood reach. Families can check these details alongside school options, budgets and daily routines.

Trying different priorities

Setting aside Oslo's work-commute proxy selects Zurich under both weight sets. In hypothetical ten-per-cent changes, baseline weights select Oslo in all three scenarios. Equal weights select Oslo for resource expansion and balanced improvement, and Zurich under living pressure. These are conditional tests of the model, not forecasts or a city's real chance of being best. The comparison is most useful as a prompt to identify which priorities and local facts matter for a particular student.

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Report on Use of AI Tools

Documentation status: incomplete review record. This appendix records assistance supported by saved task metadata and research receipts. The complete team AI history remains to be reconciled with actual saved conversations and other tool records before submission.

Known tool and scope

OpenAI Codex, used 8–11 November 2026 (Singapore time). The session header records Desktop version `0.166.0-alpha.5.1`; three inspected windows record `gpt-6.1-sol` in 74, 42 and 51 root contexts. The two continuation windows record 148 subagent spawn requests: 77 specify `gpt-6.1-terra`, 28 specify `gpt-6-luna`, and 43 omit a model or use null. Recorded identifiers and requested labels do not attest serving versions or every assistant's model. The assistant supported official-source search and extraction, data-scope checks, Python calculations and audits, LaTeX drafting, and wording revisions. Concurrent assistant tasks prepared bounded evidence checks, a controlled model diagnostic and the news-article contribution [19]. The anti-defensive-writing skill supplied editorial instructions; it is a workflow resource, not a surveyed student preference or a data source.

Exa, used 8–9 November 2026 through the recorded `web_fetch_exa` tool, retrieved and extracted text from official webpages, PDFs and data interfaces [20]. Saved receipts identify the service, source URLs and returned text, but do not specify an internal model or service version. They do not establish whether a generative model participated in every extraction. Extracted representations are distinguished from original downloaded bytes. For example, Tokyo sports-facility PDF text with ambiguous row order was excluded; the admitted five-venue lower bound uses separately downloaded Nerima municipal HTML.

Recovered records and coverage

The supporting inventory authenticates a complete-line prefix of this root task log by byte count and SHA-256. It ends at 14:31:18.850 UTC on 9 November 2026 (22:31 in Singapore). A private export retains 326 selected visible text records: six selected genuine requests, 259 progress messages and 61 complete final responses. The original 125-record selection and its private byte prefix are preserved.

The original tool-input export retains 1,865 plaintext inputs: 1,749 custom-call strings and 116 function-call argument strings, including generated code and LaTeX where present. Recorded inputs do not prove successful execution.

A continuation of the same root log ends at 07:26:15.500 UTC on 10 November 2026. It retains 197 selected visible records and 1,372 top-level plaintext assistant tool-call inputs. A third window ends at 20:32:42.716 UTC that day (04:32 on 11 November in Singapore), adding 239 visible records: three requests or replies, 191 progress messages and 45 final responses; it also retains 1,367 tool-call inputs. Across the three windows, totals are 762 selected visible records and 4,604 inputs. Nested executions, subagent sessions and later work are outside these selections.

All three windows exclude reasoning, system/developer records, tool results, attachments, subagent sessions, other chats and team members. The original log and selected exports remain outside Git (directory mode 700, file mode 600); these selections are not submission transcripts. Tool inputs remain unredacted and may contain copied content. Public inventories retain metadata and hashes; older selectors contain request prefixes, while the latest uses fingerprints. Record exclusion does not redact content, and fingerprints do not guarantee anonymity.

Verification associated with this draft

The research calculations preserve the original city CSV and the V204 primary evidence. The V204 validation independently recomputes weighted norms and self-excluded competitor maxima, verifies interval membership of constructed completions and checks source hashes. Original provider records are kept separately from derived tables. Compilation and rendered-page review check the presentation. These checks establish the specified calculation and document properties; they do not replace the team's responsibility to review factual claims, original supplied data and final authorship.

Records required to finalize this report

Finalization requires the team's actual AI tools, available models or versions, dates and purposes, with associated text and references, and saved query/response records according to the applicable COMAP policy and its exceptions. Other members' tools, translation use and earlier conversations remain unconfirmed. The registered Team Control Number and confirmation of complete team-use coverage are also required.